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ORIGINAL PAPER

Joint optimization of SNR and motion energy consumption for UAV-enabled collaborative beamforming

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Abstract
Unmanned aerial vehicles (UAVs) are usually resource constrained, and have the limited communication and energy storage capacity. Collaborative beamforming (CB) in UAV networks based on a virtual node antenna array (VNAA) can enhance the signal-to-noise-ratio (SNR) and energy efficiency of a single UAV node. The UAV nodes can move to better locations for constructing the VNAA to achieve a maximum SNR of CB. However, this will result in an extra motion energy consumption. In this paper, we formulate a joint optimization problem to simultaneously optimize the received SNR and motion energy consumption for UAV-enabled CB. Then, a mended particle swarm optimization with weed optimization mechanism algorithm is proposed to solve the formulated joint optimization problem. Simulation results verify the effectiveness of the proposed algorithm.

Keywords Unmanned aerial vehicle · Antenna array · Collaborative beamforming · Particle swarm optimization

1 Introduction

Unmanned aerial vehicle (UAV) networks can support high rate transmissions and facilitate wireless broadcast. Thus, they have been widely applied in wireless communications [1]. For example, in abominable conditions, the UAV can be used as a relay node to establish communication with the ground base station (BS), so as to quickly renovate the damaged network. In UAV networks, a UAV can be employed as the flight base station to supply reliable, economical and efficient communication service. For example, in the Olympic Games, the demand for communications is very large, and it is necessary to improve the coverage and capacity of the limited existing cellular network infrastructure. Due to the mobility and flexibility of the UAVs, they can provide reliable, cost-effective and high quality wireless communication for users. However, owing to the size and weight constrains of UAVs, the UAV networks are resource constrained and have limited communication and energy storage capacity [2].

Collaborative beamforming (CB) is a promising technique inspired by antenna arrays and it has been applied in the wireless networks to solve the abovementioned problem [3]. In UAV networks, the nodes participating in the CB can compose a virtual node antenna array (VNAA) to provide communication service for the remote BS, and the communication range, transmission performance and energy efficiency of a single UAV can be effectively enhanced. However, since the UAVs in the networks are often randomly located, achieving an optimal beam pattern performance of CB is difficult, causing the lower signal-to-noise ratio (SNR). To achieve a better communication performance, the UAVs can move to better locations for performing CB. However, there is no doubt that extra motion energy of UAVs is consumed. Thus, how to determine the optimal locations of UAV elements so that achieving higher SNR while saving motion energy consumption of UAVs is a key issue for utilizing CB in UAV networks.

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There has been some research on the communication performance in UAV networks. Wu et al. [4] propose an algorithm to optimize the trajectories of UAVs and control the power in multi-UAV enabled wireless networks at the same time. He et al. [5] jointly optimize the flying altitude of the UAVs and antenna beamwidth for the throughput optimization. Zeng et al. [6] develop a new solution to optimize flight path of UAVs, and this method maximizes the energy efficiency of the UAV communication. Saad et al. [7] use a deployment approach to design the locations of the UAVs and this achieves a minimum power consumption in the UAVs. Li et al. [8] propose an optimal transmission schedule of the UAVs to make the energy consumption decrease. Sambo et al. [9] design the trajectories of the UAVs in order to provide communication service with consuming the least energy. Moreover, CB has been successfully applied in conventional wireless networks. For example, the authors in [10] analyze the performance of CB in wireless ad hoc sensor networks by using the theory of random arrays. Ahmed et al. [11] adopt Gaussian probability density function (PDF) to optimize the locations of nodes participating in CB in wireless sensor networks (WSN) and the CB achieves better performance compared to the case of uniform PDF. Nijsure et al. [12] use CB to achieve the ground-to-air transmissions in WSN-UAV monitoring system, and the experimental results indicate that CB achieves better performance for long distance transmission than direct transmission. Vorobyov et al. [13] propose a strategy to achieve power control at each node in CB and the CB with power control has better performance compared to the CB with equal power. Garza et al. [2] propose the design of UAV-enabled three-dimensional (3D) antenna arrays to maximize the performance according to the sidelobe level (SLL) and directivity, and the experimental results show that the optimized design enlarges the range of operation for UAV communications and improves performance in coordinated flight formations. Mozaffari et al. [14] develop a novel method to employ a UAV-enabled antenna array system. The proposed drone antenna arrays make progress in the spectral, efficiency and service time. Moreover, the appropriate spacing of array element can increase the gain of UAV-enabled antenna arrays. Different from the previous works that aim to optimize the communication performance and energy control of UAVs separately, we realize the joint optimization of improving the SNR and cutting down the motion energy consumption for UAV-enabled CB.

The main contributions of this paper are summarized as follows:

(1) We formulate a joint optimization problem to simultaneously increase the received SNR and reduce motion energy consumption for UAV-enabled CB.
(2) We adopt a particle swarm optimization with weed optimization mechanism (PSOWOM) algorithm to solve the formulated problem. PSOWOM adopts the weed optimization mechanism (WOM) to make it more suitable for the formulated optimization problem.
(3) We adopt several simulations to prove the performance of the proposed algorithm. Several classical evolutionary computation algorithms are used for comparison, to assess the performance of the proposed PSOWOM for solving the formulated problem. Moreover, we test the stability of the proposed algorithm. In addition, we verify the effectiveness of the improved mechanism.

The rest of this paper is organized as follows. Section 2 reviews some key related works. Section 3 presents the system model and formulates the joint optimization problem. Section 4 proposes the PSOWOM algorithm, and Sect. 5 shows the experimental results. Finally, Sect. 6 concludes the overall paper.

2 Related work

In this work, we aim to achieve joint optimization of SNR and motion energy consumption for UAV-enabled CB. However, there are some literatures that consider these objectives separately in recent years, and we offer a brief description of some key works of them to illustrate the advantages of the proposed work.

Kumar et al. [15] present a survey of machine learning algorithms for WSN and discuss the problem of energy harvesting that can be solvable using machine learning approaches in this network. Sun et al. [16] study the UAV networks based on CB and formulate a time and energy minimization communication multi-objective optimization problem by simultaneously minimizing the total transmission time, total performing time of UAV-enabled virtual antenna arrays and total energy consumptions of UAVs. Hsu et al. [17] propose a reinforcement learning approach of collision avoidance and study optimal trajectory planning for UAV communication networks. They help UAVs learn collision avoidance by using reinforcement learning and developing a shortest backward path to collect data from other devices. However, the motion energy consumption of UAVs is not considered in this work.

Wu et al. [18] investigate the UAV-enabled uplink communication and formulate a throughput maximization problem solved by applying the block coordinate descent and successive convex optimization techniques. Garg [19] studies the multi-

UAV system where the UAVs are used as mobile base stations and proposes an iterative approach to maximize throughput by jointly optimizing transmission power, UAV trajectories and user scheduling. Liu et al. [20] adopt a novel framework for multi-UAV assisted wireless networks. They develop the multi-agent Q-learning-based trajectory design and the power control algorithm to maximize the instantaneous sum transmit rate while satisfying the rate requirements of users. Zhao et al. [21] utilize a deep deterministic policy gradient approach to achieve the joint optimization of trajectory design and power allocation in UAV networks and the proposed method has the best performance on overall network utility with faster processing speed and higher system capacity. Sun et al. [22] investigate the UAV-enabled wireless communication network and formulate a total energy consumption minimization problem by jointly optimizing the UAV trajectory, transmit power and time slot. Xu et al. [23] integrate Licensed-assisted Access technology into UAV networks and develop a joint optimization algorithm of trajectory design and transmission to maximize the energy efficiency of the UAV, under which the energy efficiency is significantly enhanced by UAV mobility. Gao et al. [24] propose a novel energy-saving design for robust secure transmission in UAV networks and this method minimizes the energy of UAV by jointly optimizing the trajectories of UAVs and user scheduling.

These abovementioned works are only optimized from one aspect of communication performance of the UAV networks or energy consumption control of UAVs. However, the communication performance of the UAV networks is limited due to the limited energy of the UAV. The joint optimization of the communication performance and motion energy consumption of UAVs is not considered in the previous works.

Different from the abovementioned works, we consider the joint optimization of SNR and motion energy consumption for UAV-enabled CB in this paper. CB in UAV networks based on a VNAA can increase the SNR and energy efficiency of a single UAV node. SNR is an important index to describe the gain of VNAA. Increasing SNR can improve the gain, so as to enhance the communication performance of the UAV networks. The locations of the UAVs is the key factor that affects the SNR. UAVs can move to the optimal locations to obtain the maximum SNR, but this will increase the motion energy consumption of UAVs. Therefore, we aim to solve the joint optimization problem of SNR and motion energy consumption for UAV-enabled CB.

3 System model and problem formulation

In this section, we introduce the system model and energy consumption model of the UAV, and formulate the joint optimization problem.

3.1 Geometry of UAV-enabled VNAA

We consider an air-to-ground (A2G) communication system consisting of a UAV-enabled VNAA, as shown in Fig. 1. Specifically, a single omnidirectional antenna is equipped in each UAV. Without loss of generality, we use a 3D Cartesian coordinate system in the work, and the location of the ith UAV is denoted as (x_i^U, y_i^U, z_i^U). On the basis of the principle of electromagnetic wave superposition, the array factor (AF) of the UAV-enabled VNAA can be expressed as follows [14]:

$$AF(\theta,\phi) = \sum_{i=1}^{N} I_i e^{j\left[k_c\left(x_i^U \sin\theta\cos\phi + y_i^U \sin\theta\sin\phi + z_i^U \cos\theta\right)\right]} \quad (1)$$

where $\phi \in [-\pi, \pi]$ and $\theta \in [0, \pi]$ denote the azimuth angles and elevation, respectively. Moreover, N denotes the number of UAVs, I_i denotes the excitation current weight of the ith UAV, λ denotes the wavelength, and $k_c = 2\pi/\lambda$ denotes the phase constant.

3.2 Received SNR of A2G system

The received SNR of the CB-based A2G system can be described as follows [14] [25]:

$$SNR = \frac{r^{-\alpha} P_t K_O G(x,y,z)}{\sigma^2} \quad (2)$$

$$G(x,y,z) = \frac{4\pi |AF(\theta_0,\phi_0)|^2 \omega(\theta_0,\phi_0)^2}{\int_0^{2\pi}\int_0^{\pi} |AF(\theta,\phi)|^2 \omega(\theta,\phi)^2 \sin\theta d\theta d\phi}\eta \quad (3)$$

where $G(x,y,z)$ denotes the gain of the VNAA towards the BS, wherein $x = [x_i^U]$, $y = [y_i^U]$,$z = [z_i^U]$. r denotes the distance between the BS and the origin of the array, α is the pathloss exponent, P_t is the whole transmit power of the array, K_O is the constant path loss coefficient, σ^2 is the noise power, and $\omega(\theta,\phi)$ is the magnitude of the far-field radiation pattern of each element.

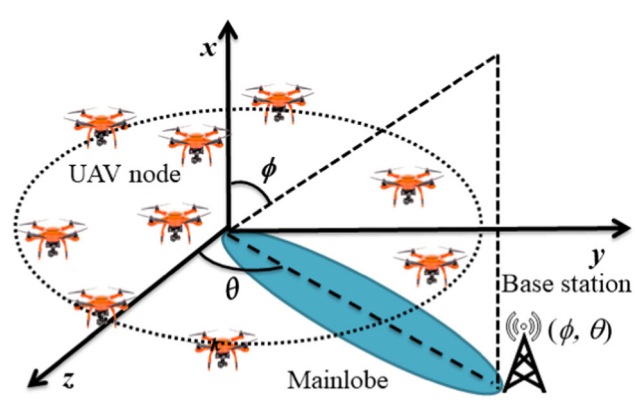

Fig. 1 Geometry of UAV-enabled VNAA.

3.3 Energy consumption model of the UAV

The energy consumption of a UAV is usually used for propulsion and communication. The propulsion energy of a UAV usually far exceeds the communication energy [26]. Therefore, the communication energy is negligible in the work, and the propulsion power consumption is expressed as follows [27]:

$$P(v) = \underbrace{P_0\left(1+\frac{3v^2}{v_{tip}^2}\right)}_{\textbf{blade profile}} + \underbrace{P_i\left(\sqrt{1+\frac{v^4}{4v_0^4}}-\frac{v^2}{2v_0^2}\right)^{1/2}}_{\textbf{induced}} + \underbrace{\frac{1}{2}d_0\rho s A v^3}_{\textbf{parasite}} \tag{4}$$

where P_i and P_0 denote the induced power in hovering status and blade profile power, respectively, v_0 denotes the average rotor induced velocity in hover, v_{tip}^2 denotes the tip speed of the rotor blade, ρ is the air density, d_0 is the fuselage drag ratio, A is the rotor disc area, s is the rotor solidity, D is the flight length of the UAV, T denotes the given finite time for the propulsion of UAVs, and $v = \frac{D}{T}$ denotes the speed of a UAV.

3.4 Problem formulation

In this work, a UAV-enabled wireless communication system is considered, where a set of rotary-wing UAVs are located in a 3D monitor area to communicate with a remote BS. When the collected data needs to be transmitted to the BS, these UAVs will form a VNAA and transmit the signal through CB. Hence, these UAVs need to fly to new locations to form VNAA for CB.

As mentioned above, the UAVs that are randomly deployed may damage the beam pattern of the VNAA, so that further reducing the received SNR at the BS. Thus, it is necessary for the UAVs to fly to better locations to achieve a better beam pattern as well as the received SNR at the BS. However, this will increase the motion energy consumption of UAVs. Moreover, the energy of a UAV is limited. Thus, suitable locations of UAVs need to be determined so that jointly optimizing the received SNR at the BS and motion energy consumption of UAVs. Therefore, the joint optimization problem of SNR and motion energy consumption for UAV-enabled CB can be formulated as follows:

$$\begin{aligned} \min_{x_i^U, y_i^U, z_i^U} \quad & F = w_1 \cdot \frac{1}{SNR} + w_2 \cdot P(v)T \\ \text{s.t.} \quad & 0 \leqslant x_i^U \leqslant l \\ & 0 \leqslant y_i^U \leqslant w \\ & 0 \leqslant z_i^U \leqslant h \end{aligned} \tag{5}$$

where w_1 and w_2 denote weighting factors on the basis of various conditions. Moreover, l, h and w are the limited length, height and width of the range of the A2G communication system, respectively.

4 Algorithm

Particle swarm optimization (PSO) algorithm is a popular method for solving the optimization problems [28]. However, conventional PSO may be not efficient for the formulated joint optimization problem for UAV-enabled CB. This may be because that the optimization problem is too complex to obtain the best solution. In addition, the conventional PSO has some drawbacks such as poor global search ability. Thus, we propose the PSOWOM algorithm, which uses the WOM to enhance accuracy and global search capability of the conventional PSO for the formulated optimization problem. The conventional PSO algorithm and WOM are presented in the following parts. Moreover, we illustrate the motivation for using PSO as the basic algorithm for solving the formulated joint optimization problem. In addition, we analyze the complexity of the proposed algorithm.

4.1 Conventional PSO

PSO is a population-based evolutionary computation algorithm aroused by the groups of fish and birds searching for food and it is first developed by Kennedy and Eberhart in 1995 [29].

Moreover, it is shown that this optimizer not only is easier to implement for the less mathematical pre-processing, but also successful in a wide variety of optimization tasks [30]. In PSO, a fixed number of particles compose the entire population, and each particle includes a set of decision variables, which is defined as a solution of an optimization problem. In addition, the position and velocity are the main properties of each particle.

Each position denotes a possible solution of the optimization problem, and these particles are evaluated by a fitness function. The position of each particle tracked in the solution place is related to the personal best solution obtained so far, and this value is defined as P_{best}. The position in the solution space attained to the best solution

achieved by all particles is called G_{best}. Thus, the G_{best} is unique in each iteration. P_{best} and G_{best} are used to update the velocity and position of each particle. The velocity and position can be updated as follows:

$$x_i^{t+1} = x_i^t + v_i^{t+1} \tag{6}$$

$$v_{i+1}^{t+1} = \omega \cdot v_i^t + c_1 \cdot r_1 \cdot \left(P_{\text{best}_i}^t - x_i^t\right) + c_2 \cdot r_2 \cdot \left(G_{\text{best}} - x_i^t\right) \tag{7}$$

where x_t^i and v_t^i represent the position and velocity of the ith particle in the tth generation, respectively, c_1 and c_2 represent accelerating coefficients, and they range from [0, 1]. ω represents the inertia coefficient, and it is in the range of [0, 1]. Furthermore, r_1 and r_2 represent random digits between [0, 1].

The Algorithm 1 shows the pseudo code of the conventional PSO, and the main calculation process of the conventional PSO is as follows [31]:

Step 1: Definition. Define fitness function $f(x)$, solution space, and population size N.

Step 2: Initialization. Initialize the population with random position and velocity.

Step 3: Evaluate the fitness function value (FFV) of every particle in the M-dimensional solution space.

Step 4: Update the P_{best} and G_{best} based on the FFV. First, for each particle position $x_k, k \in (1, ..., M)$, compare the FFV of x_k with P_{best_k}. If $f(x_k)$ is better than $f(P_{best_k})$, then set $f(P_{best_k})$ equal to the new $f(x_k)$, and set the position of P_{best} equal to the current position of x_k; otherwise, the current P_{best} is reserved. Second, compare $f(P_{best_k})$ with $f(G_{best})$. If $f(P_{best_k})$ is better than $f(G_{best})$, then set G_{best} equal to the new P_{best_k}; otherwise, the current G_{best} is reserved.

Step 5: Update the position and velocity of each particle according to Eqs. (6) and (7).

Step 6: Return to step 3 until the maximum value of iteration is satisfied.

Algorithm 1: Conventional PSO

```
Initialize a population of N particles;
Initialize the maximum value of iter_max iterations;
Initialize the position and velocity of every particle;
Calculate the FFV of each particle;
for t = 1 to iter_max do
    for i = 1 to N do
        Update each particle using Eqs. (6) and (7);
        Calculate the FFVs of the particles;
        Update the P_best and G_best based on the FFV;
    end
end
Return G_best.
```

4.2 Motivation for using PSO

In this work, the formulated problem to simultaneously increase the received SNR and reduce motion energy consumption of the UAVs for CB is a complex joint optimization problem and requires high computational complexity. First, the positions of the UAVs affect the SNR and motion energy consumption for the UAV-enabled CB. The UAV nodes can move to better locations for constructing the VNAA to obtain a maximum SNR of CB. However, the motion energy consumption of the UAVs will be increased. Thus, these optimization objectives are trade-offs. Second, as mentioned in Sect. 3, we consider an A2G communication system consisting of a UAV-enabled VNAA and formulate a joint optimization problem. In other words, the UAV positions are the solution of the formulated problem.

However, the formulated problem is NP-hard, and the reasons are as follows. Beam pattern optimization is an NP hard problem [32]. As shown in Eq. (2), SNR can be specified as the beam pattern optimization, and increasing the SNR is one of the objectives of the formulated joint optimization problem. Thus, the formulated joint optimization problem is also an NP hard problem, which means that finding the optimal solution to solve the formulated joint optimization problem in polynomial time cannot be achieved. Therefore, one reasonable method is to use the evolutionary algorithms to seek a feasible solution in finite iterations, which is practical for dealing with the considered problem in this work. Moreover, another advantage of evolutionary algorithms is that they can solve the problem directly without considering the constraints of the optimization problems [33]. Compared to other evolutionary algorithms, PSO has its advantages, e.g., efficient implementation and fast convergence [34]. Moreover, PSO has a memory function, which means that the best historical position in the particle population can be recorded, and the information can be shared between these solutions in the population. In addition, the inertia property of particles helps the algorithm find the best solution effectively [35]. Each particle moves with a fixed amount of inertia in the search space, while being attracted to the optimal position that they have found independently, and to the best position found by any particle in its neighborhood. The momentum provided by the inertia helps the particles search the solution space thoroughly, which means that the PSO algorithm avoids stopping prematurely and falling into local optima. Thus, these characteristics motivate us to use PSO as the basic algorithm for solving the formulated joint optimization problem.

4.3 WOM

WOM comes from the invasive weed optimization (IWO) algorithm [36], which is composed of three stages: reproduction, spatial dispersal, and competitive exclusion. The process of WOM is as follows [37]:

Step 1: Sort the FFV of each particle.

Step 2: Reproduction. Each particle generates seeds based on the FFV according to Eq. (8).

$$S_{num} = floor\left[S_{\min} + (S_{\max} - S_{\min}) \times \frac{f - f_{worst}}{f_{best} - f_{worst}}\right] \tag{8}$$

where floor represents the round down operation, $S_{\max}$ and $S_{\min}$ represent the maximum and minimum value of seeds, respectively, f denotes the $f(P_{best_k})$ of each particle. Moreover, f_{best} and f_{worst} represent the best and worst FFVs in the current iteration, respectively.

Step 3: Spatial dispersal. In the M-dimensional solution space, there are new particles and original particles. Eq. (9) shows the standard deviation in this procedure, and the generation of the new particles is shown in Eq. (10).

$$\sigma_{iter} = \frac{(iter_{\max} - iter)^m}{(iter_{\max} - 1)^m} \times (\sigma_{initial} - \sigma_{final}) + \sigma_{final} \tag{9}$$

$$W_{new}^{iter+1} = P^{iter} + N(0, \sigma_{iter}^2) \tag{10}$$

where $\sigma_{initial}$ and σ_{final} are the initialized initial and final standard deviations, respectively, m is the nonlinear modulation index, $iter$ denotes the current iteration, $iter_{max}$ is the maximum iteration, P^{iter} denotes the individual representing the original particles in the $iter$th iteration, W_{new}^{iter+1} denotes a new particle produced from the original particle in the $(iter + 1)$ iteration, and $N(0, \sigma_{iter}^2)$ denotes a random number produced from the normal distribution with mean zero and standard deviation (SD).

Step 4: Competitive exclusion. The sum of particles will excess the limited maximum number of particles after reproduction. Thus, particles with worse FFVs will be eliminated to keep the maximum number of population. Then, the preserved particles will update their velocities and positions.

4.4 PSOWOM

Algorithm 2 shows the pseudo code of the PSOWOM, and the process is as follows.

First, PSOWOM initializes all parameters (e.g., population size), position and velocity of every particle. Second, the candidate solutions are updated by Eqs. (6) and (7). Third, the obtained solutions will generate new seeds, furthermore, the new particles will be obtained by Eq. (10). Moreover, the P_{best} and G_{best} are updated based on the FFV. Finally, particles with worse FFVs will be eliminated to keep the maximum number of population. This evolutional process will be performed iteratively until the maximum value $iter_{max}$ is reached.

Algorithm 2: PSOWOM

```
Initialize a population of N particles;
Initialize the maximum number of iter_max iterations;
Initialize the position and velocity of each particle;
Calculate the FFV of each particle;
for t = 1 to iter_max do
    Select the f_worst and f_best in the population by sorting;
    for i = 1 to N do
        Renew each particle using Eqs. (6) and (7) ;
        Produce new seeds: calculate S_num in the reproduction using Eq. (8);
        Distribute the seeds in the spatial distribution to obtain new particles by using Eq. (10);
        Calculate the FFVs of the particles;
        Update the P_best and G_best based on the FFV;
    end
    Eliminate individuals with higher FFVs in the colony to reach N;
end
Return G_best.
```

4.5 Complexity of the proposed PSOWOM

In this section, we analyze the time complexity of the proposed PSOWOM. The most time-consuming step in the PSOWOM should be the computation of the FFV, which is several orders of magnitude more complex than the other steps. Therefore, other computation steps can be ignored. We assume that the population size and the maximum iteration are N and $iter_{max}$, respectively. Thus, the complexity of PSOWOM is $O(N * iter_{max})$ because the algorithm calculates the FFV for $N * iter_{max}$ times, which is similar to the conventional PSO. However, PSOWOM may take longer computing time than conventional PSO in practical situation, and the reason may be that the introduced improved mechanism results in extra computing time, which is difficult to be predicted. Therefore, to assess the time complexity of the proposed PSOWOM more comprehensively, the CPU time of the proposed PSOWOM is calculated in Sect. 5.

4.6 Joint optimization of SNR and motion energy consumption for UAV-enabled CB using PSOWOM

The UAVs are considered as the antenna elements. We use the proposed PSOWOM algorithm to solve the joint optimization problem of the SNR and motion energy consumption for UAV-enabled CB by optimizing the location of each UAV, and consider the solution (particles) of the algorithm as the locations of the UAVs. The solutions of the formulated problem can be expressed in Eq. 11.

$$P = (x_1, x_2, x_3, \cdots, x_n, y_1, y_2, y_3, \cdots, y_n, z_1, z_2, z_3, \cdots, z_n) \tag{11}$$

where P is the particle of the proposed PSOWOM algorithm, n denotes the amount of the antenna elements. Moreover, the population of PSOWOM is shown in Eq. 12. Figure 2 shows the process of solving the formulated joint optimization problem using the proposed PSOWOM. First, we initialize the population of N particles which are candidate solutions, and calculate the FFVs of them. Second, the population is updated by using the WOM to obtain new population. Third, the number of iteration is checked to see if the maximum iteration is reached. If it is reached, the one with the minimum FFV is selected as the solution; otherwise continue to calculate the FFV of each particle and update the candidate solutions.

$$pop = \begin{bmatrix} P_1 \\ P_2 \\ P_3 \\ \cdots \\ P_N \end{bmatrix} = \begin{bmatrix} x_1^1 & x_2^1 & x_3^1 & \cdots & x_n^1 & y_1^1 & y_2^1 & y_3^1 \\ \cdots & y_n^1 & z_1^1 & z_2^1 & z_3^1 & \cdots & z_n^1 & \\ x_1^2 & x_2^2 & x_3^2 & \cdots & x_n^2 & y_1^2 & y_2^2 & y_3^2 \\ \cdots & y_n^2 & z_1^2 & z_2^2 & z_3^2 & \cdots & z_n^2 & \\ & & \cdots & & & & & \\ x_1^N & x_2^N & x_3^N & \cdots & x_n^N & y_1^N & y_2^N & y_3^N \\ \cdots & y_n^N & z_1^N & z_2^N & z_3^N & \cdots & z_n^N & \end{bmatrix} \tag{12}$$

where N denotes the population size.

5 Simulation results

In this section, simulations are conducted to evaluate the performance of the proposed PSOWOM for solving the formulated joint optimization problem shown in Eq. 5 by using MATLAB. The computer used for the simulation is with an 8 GB RAM, a Windows 10 operating system and an Intel (R) Core (TM) i3-8100 CPU. In the simulation, we consider two cases with different numbers of UAVs (16 UAVs for Case 1 and 32 UAVs for Case 2), and the monitor region is set as 100 m × 100 m. Moreover, the minimum and maximum altitudes of UAVs are set as 100 m and 120 m, respectively. In addition, the population size and maximum number of iterations of the algorithms are 20 and 200, respectively. Table 1 shows the main simulation parameters.

We use the proposed PSOWOM to solve the formulated joint optimization problem, and the experimental results are compared to conventional PSO [38], biogeography-based optimization (BBO) [39], cuckoo search (CS) [40], firefly algorithm (FA) [41] and IWO [36], respectively. Moreover, we compare the results optimized by mentioned algorithms with the results obtained by linear antenna array (LAA) and random antenna array (RAA). Note that the LAA and RAA mean that the UAVs are arranged in a line and randomly arranged in space, respectively.

5.1 Performance analysis of PSOWOM

We perform a comparative experiment to evaluate the overall performance of the proposed PSOWOM algorithm before solving the formulated joint optimization problem of SNR and motion energy consumption for UAV-enabled CB. We use the PSOWOM and abovementioned algorithms to acquire the minimum values of nine selected functions (i.e., F_3, F_6, F_{11}, F_{17}, F_{18}, F_{20}, F_{21}, F_{23}, and F_{27}) of the standard CEC 2014 function set [42]. We set the dimension of the population size, the solution, and the maximum number of iterations of every algorithm to 20, 10, and 1000, respectively. Furthermore, we make the tests run independently for 30 times on each function, and Table 2 shows the statistical consequences. Note that the values with bold entries in this table represent the best values obtained by various algorithms. As can be seen, the proposed PSOWOM algorithm achieves the better performance for seven of the nine functions compared to other algorithms. In addition, the t-test [43] experiment is

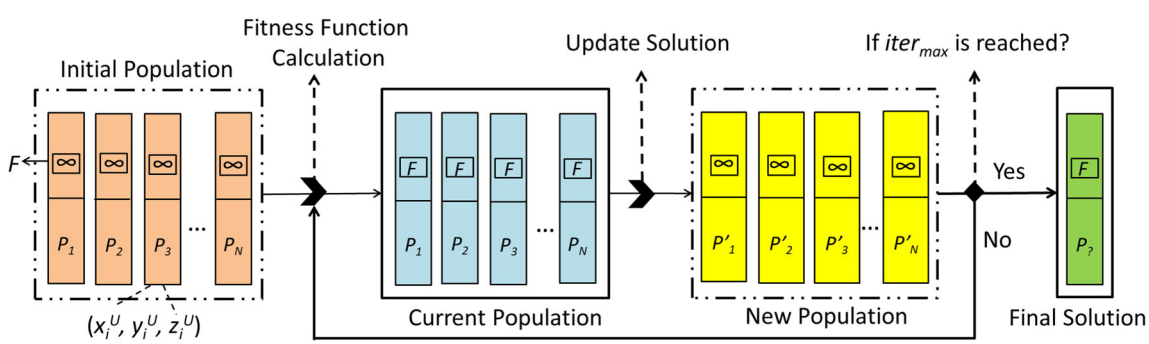

Fig. 2 Diagrammatic sketch of solving the formulated joint optimization problem using the proposed PSOWOM.

Table 1 Main simulation parameters

Parameter	Description	Value
v_0	Average rotor induced velocity	4.03
v_{tip}	Tip speed	120
ρ	Air density	1.225
$d0$	Fuselage drag ratio	0.6
A	Rotor disc area	0.503
s	Rotor solidity	0.05
δ	Profile drag coefficient	0.012
ω	Blade angular	300

conducted to statistically analyze the comparison results of the Table 2. The numerical results of t-test of various algorithms are listed in Table 3. Note that '1' and '-1' represent the results that if PSOWOM is better or worse than the comparison algorithms, respectively, and the other results are denoted by '0'. Moreover, if the number of the function value of PSOWOM is smaller or equal and even no domination than it of comparison algorithms, we will define it as S, E, and L, respectively. In addition, the $S-L$ line shows the differences obtained by the amount of S and L of PSOWOM. As can be seen, the numbers of results that the PSOWOM outperforms BBO, CS, FA, IWO, and PSO are 8, 9, 8, 9 and 9 in nine functions, respectively. Correspondingly, the number of having no significant difference is all 0. Note that the larger difference in the $S-L$ line means the superior effectiveness of PSOWOM. Therefore, the results show that the proposed PSOWOM algorithm can be expected to become a trustworthy solution for optimization problems.

5.2 Tuning of parameter values

The optimal values of parameters in an optimization algorithm are various when solving different optimization problems. Therefore, we tune the key parameters of the proposed PSOWOM algorithm so that making it obtain the best result for the formulated joint optimization problem. In the proposed PSOWOM, m is one of the key parameters, which influences the diversity of the population in the WOM. In the process of tuning, PSOWOM is used to obtain the solution for the formulated optimization problem by changing the values of m from 1 to 10, and the step size is set to be 1. We repeat the test for 30 times to avoid randomness, and the mean values are presented in Fig. 3. As can be seen, the optimal value of m is 7 for dealing with the formulated joint optimization problem.

In addition, the previous studies provide parametric values for other algorithms. For BBO [39], the emigration rate E=1, mutation rate r=0.005, and immigration rate I=1. For CS [40], the step size of Lévy flight α=1, and the probability that an egg is discovered p_a=0.25. For FA [41], the randomization parameter α=0.25, attractiveness β_0=0.2, and absorption coefficient γ=0.25. For IWO [36], the minimum number of seeds S_{min}=0, the maximum value of seeds S_{max}=5, the initial standard deviation $\sigma_{initial}$=0.01, the final standard deviation σ_{final}=0.05, and the nonlinear modulation index m=3. For PSO [38], the first learning factor $C1$=2.0, and the second learning factor $C2$=2.0.

The values of parameters used in the proposed PSOWOM and other algorithms are shown in Table 4.

5.3 Joint optimization of SNR and motion energy consumption for UAV-enabled CB

In this section, we consider 16-UAV and 32-UAV VNAAs, and the proposed PSOWOM algorithm is adopted to solve the formulated joint optimization problem of SNR and motion energy consumption for UAV-enabled CB. Moreover, the results are compared to the abovementioned algorithms.

5.3.1 Case 1: 16 UAVs

Figure 4 shows the convergence rates of PSO, BBO, FA, CS, IWO and the proposed PSOWOM, and the statistical results of joint optimization of SNR and motion energy consumption for the 16-UAV-enabled CB are shown in Table 5. Note that the values emphasized in bold font in this table represent the best values obtained by various algorithms. As can be seen, the proposed PSOWOM algorithm obtains the optimal performance in terms of convergence rate and accuracy among all the algorithms. The reason may be that the introduced WOM enhances the global search capability of the conventional PSO for solving the joint optimization problem of SNR and motion energy consumption for 16-UAV-enabled CB. Although PSO achieves the minimum energy consumption of 5.3905E+03 J, the proposed PSOWOM obtains the maximum SNR for the 16-UAV-enabled CB. This is because that these optimization objectives are trade-offs. Moreover, Fig. 5 shows the positions of UAVs before and after the optimization. The UAVs will fly to the positions optimized by the mentioned algorithms from the unified original positions. In addition, the numerical statistical results of CPU time obtained by different algorithms are shown in Table 6. Note that "Maximum", "Minimum", "Mean" and "Std." represent the maximum value, minimum value, mean value and standard deviation of the 30 independent tests, respectively. It can be seen from this table that the proposed PSOWOM takes the third most time in these

Table 2 Comparison of results of F_3, F_6, F_{11}, F_{17}, F_{18}, F_{20}, F_{21}, F_{23}, and F_{27} for 30 independent runs

ID	Algorithm	Best	Worst	Standard Deviation	Mean	Time
F_3	BBO	8.5696E+02	1.6922E+04	4.5541E+03	7.3724E+03	3.0411E+01
	CS	2.3559E+03	4.3651E+06	7.9208E+05	1.7198E+05	9.2855E−01
	FA	5.5243E+02	2.7436E+04	5.5523E+03	9.4893E+03	2.0482E+00
	IWO	1.4419E+04	1.0566E+05	2.4505E+04	5.8564E+04	7.9079E−01
	PSO	3.0935E+02	1.9749E+04	5.0261E+03	3.8870E+03	7.6507E−01
	PSOWOM	5.0927E+02	5.7805E+03	1.4226E+03	**2.1166E+03**	2.5345E+00
F_6	BBO	6.0163E+02	6.0671E+02	1.2320E+00	6.0416E+02	4.5301E+01
	CS	6.0555E+02	6.1013E+02	9.3053E−01	6.0825E+02	2.0769E+00
	FA	6.0003E+02	6.0209E+02	5.9976E−01	6.0041E+02	2.3096E+00
	IWO	6.1001E+02	6.1715E+02	1.5554E+00	6.1380E+02	1.5565E+00
	PSO	6.0013E+02	6.0700E+02	1.7903E+00	6.0256E+02	1.0058E+00
	PSOWOM	6.0000E+02	6.0626E+02	1.6434E+00	6.0211E+02	3.7188E+00
F_{11}	BBO	1.3470E+03	2.2165E+03	1.8319E+02	1.6592E+03	3.1288E+01
	CS	1.7549E+03	2.8710E+03	2.5397E+02	2.3892E+03	9.5745E−01
	FA	1.1186E+03	2.2265E+03	3.0716E+02	1.7229E+03	2.0649E+00
	IWO	1.6018E+03	3.2922E+03	3.9984E+02	2.5613E+03	8.2231E−01
	PSO	1.3366E+03	2.6481E+03	3.3379E+02	1.9780E+03	7.7746E−01
	PSOWOM	1.1153E+03	2.3475E+03	2.5279E+02	**1.5488E+03**	2.2866E+00
F_{17}	BBO	2.3948E+05	2.8509E+06	6.6986E+05	1.0343E+06	3.1058E+01
	CS	7.8404E+04	8.5610E+06	1.7199E+06	1.7628E+06	9.7511E−01
	FA	2.4998E+03	1.2770E+04	3.0104E+03	5.5071E+03	2.0593E+00
	IWO	3.2504E+03	5.5035E+07	1.6079E+07	9.4757E+06	8.1521E−01
	PSO	1.9999E+03	1.4043E+04	3.3911E+03	5.4533E+03	7.9195E−01
	PSOWOM	2.3403E+03	1.0941E+04	2.4746E+03	**4.9281E+03**	2.5927E+00
F_{18}	BBO	3.4890E+04	5.9105E+06	1.4777E+06	5.9520E+05	3.0863E+01
	CS	9.0900E+03	4.2106E+06	1.0550E+06	7.3766E+05	9.3269E−01
	FA	1.9153E+03	3.6688E+04	1.0091E+04	1.2513E+04	2.0574E+00
	IWO	1.8590E+03	3.3119E+08	8.2127E+07	2.8952E+07	8.3019E−01
	PSO	2.2706E+03	2.3919E+04	6.2722E+03	9.7231E+03	8.0702E−01
	PSOWOM	1.8495E+03	3.2923E+04	9.1770E+03	**9.6719E+03**	2.5272E+00
F_{20}	BBO	2.4162E+03	2.5220E+05	7.0705E+04	8.3270E+04	3.0884E+01
	CS	3.1347E+03	2.1782E+06	5.9199E+05	3.0699E+05	9.5086E−01
	FA	2.1494E+03	1.5515E+04	4.1664E+03	5.6203E+03	2.0538E+00
	IWO	4.6500E+03	1.7814E+07	4.3804E+06	1.6400E+06	8.1157E−01
	PSO	2.1342E+03	1.4193E+04	3.2165E+03	5.6662E+03	7.7958E−01
	PSOWOM	2.0743E+03	1.2338E+04	2.3509E+03	**3.9698E+03**	2.5829E+00
F_{21}	BBO	3.2726E+04	1.6385E+07	5.2366E+06	4.2111E+06	3.1102E+01
	CS	3.8057E+04	3.5403E+06	1.1421E+06	1.1819E+06	9.6158E−01
	FA	2.3411E+03	1.9245E+04	4.7666E+03	7.0181E+03	2.0600E+00
	IWO	3.1611E+03	6.8308E+06	1.9332E+06	1.4253E+06	8.2229E−01
	PSO	2.2523E+03	1.2154E+04	2.4334E+03	5.3018E+03	7.8303E−01
	PSOWOM	2.3874E+03	8.8553E+03	1.4586E+03	**3.7057E+03**	2.6486E+00
F_{23}	BBO	2.5000E+03	2.5000E+03	0.0000E+00	2.5000E+03	3.4019E+01
	CS	2.5163E+03	2.6741E+03	3.0052E+01	2.6416E+03	1.1858E+00
	FA	2.6295E+03	2.6295E+03	3.5877E−06	2.6295E+03	2.1116E+00
	IWO	2.6502E+03	3.1592E+03	1.4323E+02	2.8296E+03	1.0218E+00
	PSO	2.6295E+03	2.6295E+03	3.6487E−12	2.6295E+03	8.3480E−01
	PSOWOM	2.3000E+03	2.6295E+03	6.0150E+01	2.6185E+03	3.0376E+00

Table 2 (continued)

ID	Algorithm	Best	Worst	Standard Deviation	Mean	Time
F_{27}	BBO	2.7079E+03	3.1937E+03	1.2712E+02	3.0142E+03	5.1123E+01
	CS	2.7561E+03	3.2810E+03	1.3037E+02	3.1107E+03	2.4132E+00
	FA	2.7010E+03	3.1114E+03	7.1955E+01	3.0190E+03	2.4015E+00
	IWO	3.1797E+03	3.7579E+03	1.2410E+02	3.4684E+03	1.8165E+00
	PSO	2.7023E+03	3.1353E+03	1.3397E+02	3.0299E+03	1.1311E+00
	PSOWOM	2.7023E+03	3.2077E+03	1.4584E+02	**3.0080E+03**	4.2978E+00

The values that with bold entries represent the best values obtained by various algorithms

Table 3 Numberical results of *t*-test of various algorithms

ID	BBO	CS	FA	IWO	PSO
F_3	1	1	1	1	1
F_6	1	1	– 1	1	1
F_{11}	1	1	1	1	1
F_{17}	1	1	1	1	1
F_{18}	1	1	1	1	1
F_{20}	1	1	1	1	1
F_{21}	1	1	1	1	1
F_{23}	-1	1	1	1	1
F_{27}	1	1	1	1	1
S	8	9	8	9	9
E	0	0	0	0	0
L	1	0	1	0	0
$S-L$	7	9	7	9	9

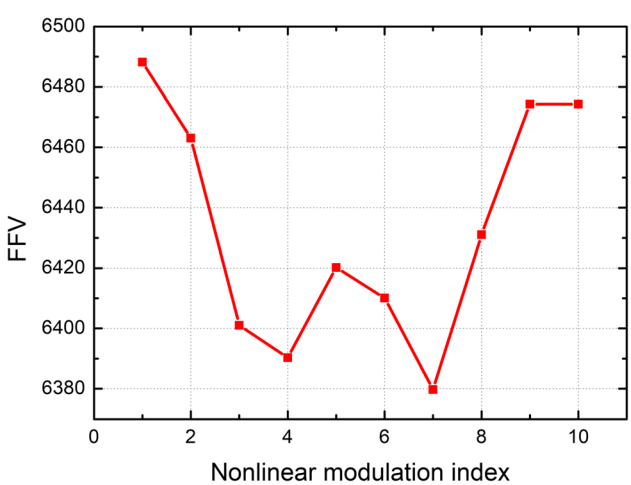

Fig. 3 Parameter tuning for m in PSOWOM for solving the formulated joint optimization problem

algorithms. The reason may be that the introduced improved mechanism of the proposed PSOWOM consumes extra computing time. The difference in CPU time between the proposed PSOWOM and other algorithms is not very large, which illustrates that the overhead of PSOWOM is similar to other algorithms. Thus, we may say that the proposed PSOWOM has the overall best performance for the joint optimization of SNR and motion energy consumption for UAV-enabled CB.

5.3.2 Case 2: 32 UAVs

Similar to the case of 16-UAV-enabled CB, we adopt the abovementioned algorithms to solve the formulated joint optimization problem of SNR and motion energy consumption for 32-UAV-enabled CB. Figure 6 shows the convergence rates obtained by various algorithms in the optimization process, and the statistical consequences are shown in Table 7. Note that the definitions of the values with bold entries in this table have the same meanings with the case 1. As can be seen, the figure and table show that the proposed PSOWOM achieves the best performance in terms of convergence rate and accuracy compared to the other algorithms. The reason may be that the reproduction and spatial dispersal process of the introduced WOM increase the diversity of the population, and avoid premature convergence. Although FA obtains the minimum energy consumption of 1.07811E+04 J, the proposed PSOWOM achieves the maximum SNR for the 32-UAV-enabled CB. The reason is that there are trade-offs between the motion energy consumption and SNR. Moreover, Table 8 shows the numerical statistical results of CPU time obtained by different algorithms. The definitions of the "Maximum", "Minimum", "Mean" and "Std." have the same meanings with the case 1. Similar to the case 1, the CPU time of the proposed PSOWOM is the second largest, but there is little difference over the other algorithms. Figure 7 shows the original positions of UAVs and the positions that UAVs need to fly for CB. In general, the proposed PSOWOM has the best performance in terms of the convergence rate and accuracy compared to the other algorithms.

Table 4 Parameter setups of different algorithms

Algorithm	Values of parameters
BBO [39]	$E = 1$, $I = 1$, $r = 0.005$
CS [40]	$\alpha = 1$, $p_a = 0.25$
FA [41]	$\alpha = 0.25$, $\beta_0 = 0.2$, $\gamma = 0.25$
IWO [36]	$S_{min} = 0$, $S_{max} = 5$, $m = 3$, $\sigma_{initial} = 0.05$, $\sigma_{final} = 0.01$
PSO [38]	$C1 = 1.5$, $C2 = 2.0$
PSOWOM	$C1 = 1.5$, $C2 = 2.0$, $\sigma_{initial} = 0.05$, $\sigma_{final} = 0.01$, $S_{min} = 0$, $m = 7$, $S_{max} = 5$

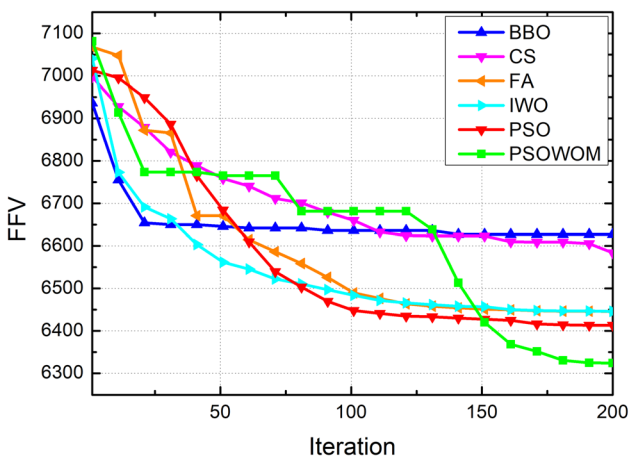

Fig. 4 Convergence rates of different algorithms for solving the formulated joint optimization problem in case 1

Table 5 Statistical results of joint optimization of SNR and motion energy consumption in case 1

Algorithm	Energy (J)	1/SNR (dB)	FFV
LAA	5.3908E+03	3.1451E−04	8.5359E+03
RAA	5.3906E+03	2.2291E−04	7.6197E+03
BBO	5.3906E+03	1.2366E−04	6.6272E+03
CS	5.3906E+03	1.1933E−04	6.5839E+03
FA	5.3908E+03	1.0548E−04	6.4457E+03
IWO	5.3908E+03	1.0552E−04	6.4460E+03
PSO	**5.3905E+03**	1.0228E−04	6.4133E+03
PSOWOM	5.3907E+03	**9.3341E−05**	**6.3241E+03**

The values that with bold entries represent the best values obtained by various algorithms

5.4 Stability test

The optimization results may be various in each time since there are randomness in evolutionary computation algorithms. Thus, it is necessary to conduct statistical tests to test the stability of the proposed PSOWOM and other algorithms for solving the formulated joint optimization problem. In this work, we use different algorithms to solve the formulated problem for 30 times, and the consequences are shown in Figs. 8 and 9, respectively, for the 16-UAV-enabled and 32-UAV-enabled CB. Moreover, the statistical results are shown in Tables. 9 and 10, respectively. As can be seen, the proposed PSOWOM is much more stable for the formulated joint optimization problem of SNR and motion energy consumption for UAV-enabled CB compared to other algorithms since it has a better performance in terms of the stability.

Table 6 Numerical statistical results of CPU time obtained by different algorithms in case 1

Algorithm	Maximum	Minimum	Mean	Std.
LAA	3.70E−01	3.30E−01	3.37E−01	7.75E−03
RAA	9.49E−01	3.60E−01	3.94E−01	1.05E−01
BBO	5.27E+04	5.23E+04	5.25E+04	1.49E+02
CS	9.88E+03	5.71E+03	7.71E+03	1.35E+03
FA	3.56E+03	1.45E+03	2.46E+03	6.63E+02
IWO	3.40E+03	2.38E+03	2.81E+03	2.87E+02
PSO	1.38E+03	6.85E+02	1.01E+03	2.19E+02
PSOWOM	4.10E+03	3.95E+03	4.02E+03	7.38E+01

5.5 Verification of effectiveness of the improved mechanism

In this section, tests are conducted to evaluate the effectiveness of the improved mechanism. Specifically, we use the conventional PSO and the PSO with WOM to solve the formulated joint optimization problem independently for 30 times, respectively, and the average value and error bar value of the FFV for case 1 and case 2 are shown in Figs. 10 and 11. As shown in the figures, PSO with the WOM can effectively improve the convergence rate and the accuracy compared to the conventional PSO for solving the joint optimization problem of SNR and motion energy consumption for 16-UAV-enabled and 32-UAV-enabled CB, respectively. In general, the PSO is efficient implementation and fast convergence, while the WOM can effectively improve the global search ability of the

Table 7 Statistical results of joint optimization of SNR and motion energy consumption in case 2

Algorithm	Energy (J)	1/SNR (dB)	FFV
LAA	1.07815E+04	2.7462E−04	1.3528E+04
RAA	1.07819E+04	1.2482E−04	1.2030E+04
BBO	1.07816E+04	6.7848E−05	1.1460E+04
CS	1.07820E+04	7.3534E−05	1.1517E+04
FA	**1.07811E+04**	6.7327E−05	1.1454E+04
IWO	1.07813E+04	5.8769E−05	1.1369E+04
PSO	1.07813E+04	7.0075E−05	1.1482E+04
PSOWOM	1.07814E+04	**5.4771E−05**	**1.1329E+04**

The values that with bold entries represent the best values obtained by various algorithms

Table 8 Numerical statistical results of CPU time obtained by different algorithms in case 2

Algorithm	Maximum	Minimum	Mean	Std.
LAA	8.64E−01	6.50E−01	6.78E−01	3.83E−02
RAA	7.50E−01	7.00E−01	7.17E−01	1.28E−02
BBO	6.96E+04	6.91E+04	6.94E+04	2.13E+02
CS	1.31E+04	1.15E+04	1.19E+04	4.77E+02
FA	2.63E+03	2.06E+03	2.27E+03	1.76E+02
IWO	1.17E+04	1.03E+04	1.06E+04	4.19E+02
PSO	3.67E+03	3.61E+03	3.65E+03	1.89E+01
PSOWOM	1.98E+04	1.88E+04	1.92E+04	3.73E+02

algorithm. Thus, the introduced improved mechanism has the ability to improve the effectiveness of the conventional PSO algorithm.

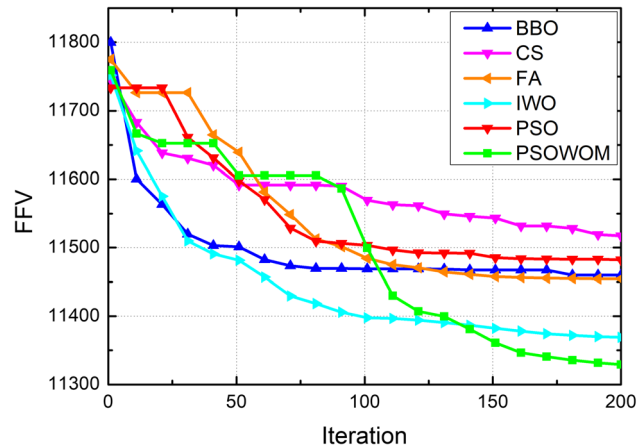

Fig. 6 Convergence rates of different algorithms for solving the formulated joint optimization problem in case 2

5.6 Limitation of the proposed PSOWOM

In this section, we analyze the limitations of the proposed PSOWOM. Although the proposed PSOWOM achieves better performance than some comparison algorithms according to the experimental results, it still has some limitations. First, the proposed PSOWOM introduces much more key parameters compared to other algorithms. This may produce difficulties in parameter tuning. Second, the proposed PSOWOM may cost longer CPU time than conventional PSO. The reason may be that the improved mechanism of PSOWOM takes more computing time. For instance, the proposed WOM needs to update the population in each loop, thus the proposed PSOWOM should take longer CPU time. Finally, we can see that the effect of the proposed PSOWOM is not significant in the early stage according to the simulation results, which means that the

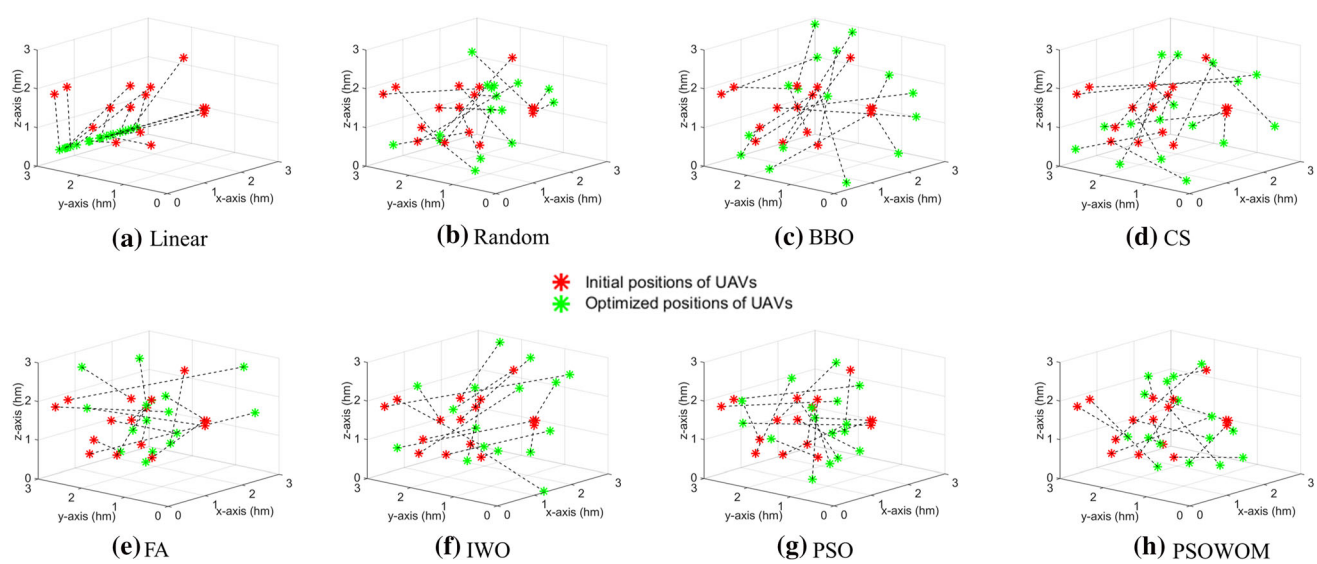

Fig. 5 Position optimization results obtained by various algorithms in case 1

Table 9 Statistical results of stability test for joint optimization of SNR and motion energy consumption in case 1

Algorithm	Best FFV	Worst FFV	Mean FFV	SD FFV
BBO	6.5030E+03	6.7012E+03	6.5758E+03	5.6658E+01
CS	6.5212E+03	6.6200E+03	6.5713E+03	2.6938E+01
FA	6.4077E+03	6.5980E+03	6.4870E+03	5.8561E+01
IWO	6.3517E+03	6.5025E+03	6.4110E+03	4.1065E+01
PSO	6.3553E+03	6.5314E+03	6.4040E+03	3.9480E+01
PSOWOM	6.3241E+03	6.3810E+03	6.3501E+03	1.6046E+01

Table 10 Statistical results of stability test for joint optimization of SNR and motion energy consumption in case 2

Algorithm	Best FFV	Worst FFV	Mean FFV	SD FFV
BBO	1.1436E+04	1.1491E+04	1.1465E+04	1.5309E+01
CS	1.1467E+04	1.1529E+04	1.1503E+04	1.6005E+01
FA	1.1412E+04	1.1528E+04	1.1469E+04	3.0835E+01
IWO	1.1343E+04	1.1468E+04	1.1378E+04	2.6692E+01
PSO	1.1366E+04	1.1529E+04	1.1436E+04	4.8781E+01
PSOWOM	1.1319E+04	1.1346E+04	1.1333E+04	7.6038E+00

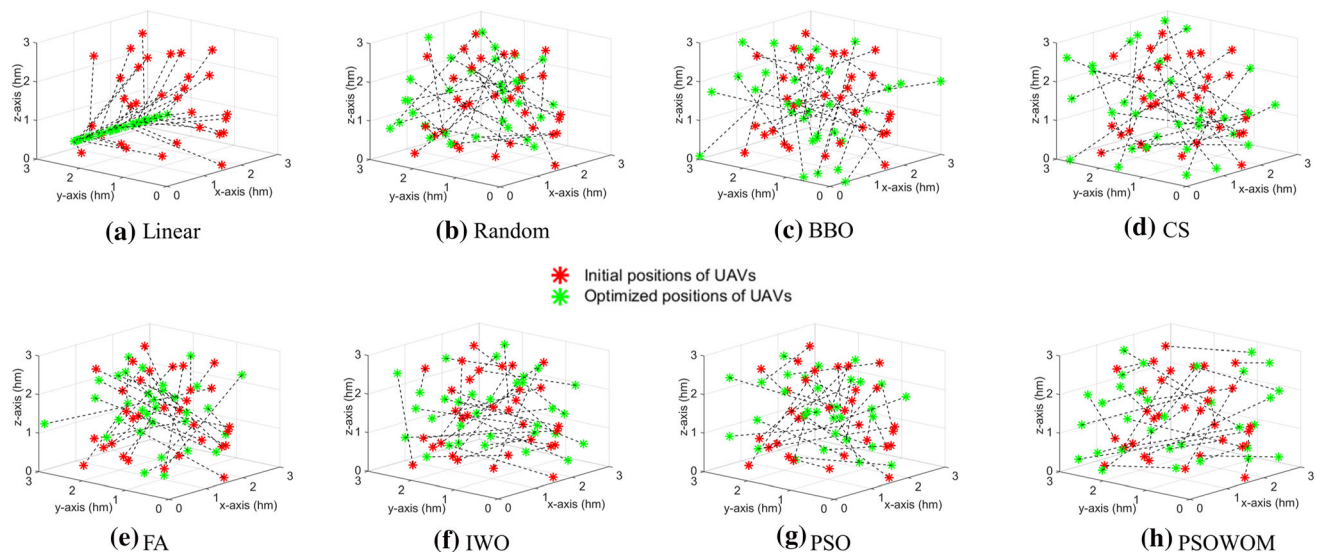

Fig. 7 Position optimization results obtained by various algorithms in case 2

algorithm needs a lot of trainings to achieve the best performance.

6 Conclusion

In this work, the joint optimization method of UAV-enabled CB is investigated. First, we formulate a joint optimization problem of SNR and motion energy consumption for UAV-enabled CB. Then, we propose a PSOWOM algorithm to solve the formulated problem. PSOWOM introduces the WOM operator to enhance the convergence rate and the accuracy of the conventional PSO algorithm for the complex optimization problem. Simulations are conducted to assess the performance of the proposed algorithm. Firstly, we adopt the CEC 2014 function set to evaluate the validity of the proposed algorithm for the optimization problem of common functions. Secondly, we tune the essential parameter of PSOWOM to make the algorithm more appropriate for solving the joint optimization problem. Thirdly, the joint optimization consequences indicate that the proposed PSOWOM algorithm obtains a better performance compared to some classical evolutionary algorithms. Fourthly, we assess the stability of the PSOWOM, and the consequences indicate that the proposed algorithm is more stable compared to other algorithms. Finally, the effectiveness of the introduced improved mechanism is verified by simulation tests, and

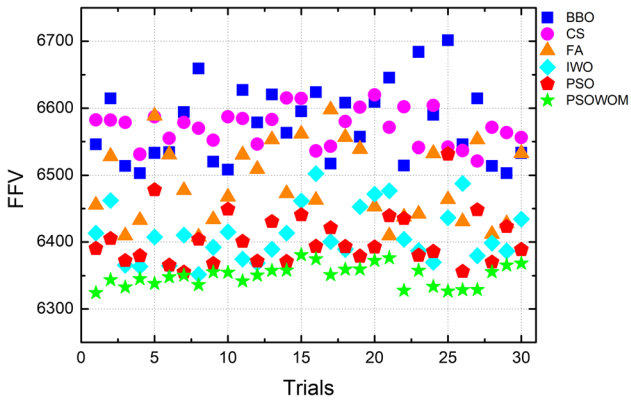

Fig. 8 Stability test results of joint optimization of SNR and motion energy consumption in case 1

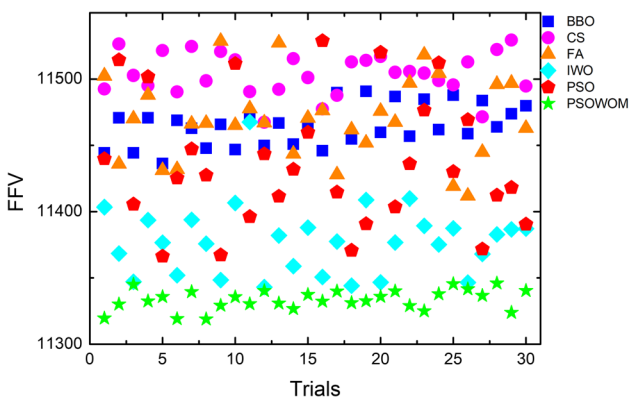

Fig. 9 Stability test results of joint optimization of SNR and motion energy consumption in case 2

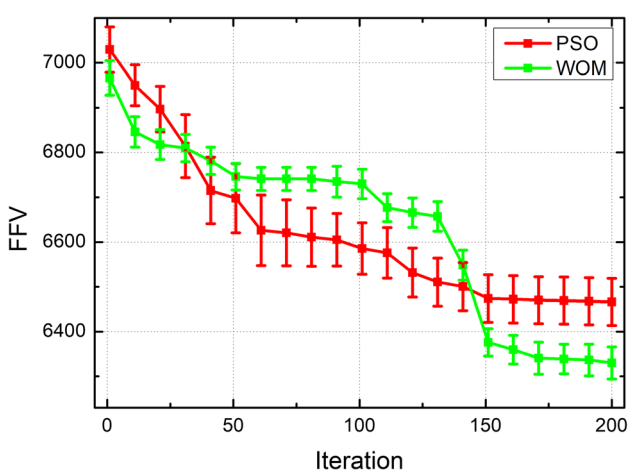

Fig. 10 Numerical results of effectiveness test of the improved mechanism in case 1

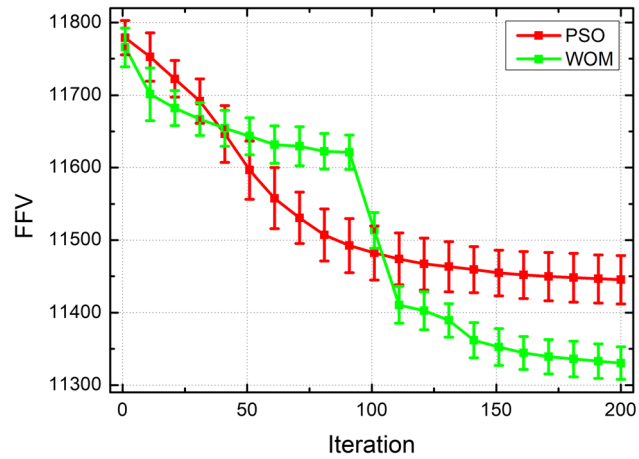

Fig. 11 Numerical results of effectiveness test of the improved mechanism in case 2

the results verify that the WOM can effectively enhance the accuracy and convergence rate of the conventional PSO.

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Data availability Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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